

Maulana Bisyr Azhari

mbazhari@kaist.ac.kr | maulanabisyira@gmail.com | [linkedin.com/in/maulanabisyirazhari](https://www.linkedin.com/in/maulanabisyirazhari) | github.com/maulanaazhari

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. in Electrical Engineering

Daejeon, South Korea

Aug. 2021 – Present

- **Dissertation:** "State Estimation for Autonomous Drone Racing and Visual Odometry Leveraging Foundation Model".
- **Advisor:** Prof. David Hyunchul Shim

University of Indonesia

B.Eng. in Electrical Engineering

Depok, Indonesia

Aug. 2016 – Jul. 2020

- **Thesis:** "Missile Dynamical System Identification using Artificial Neural Networks".
- **Advisor:** Prof. Benyamin Kusumoputro

SELECTED PUBLICATIONS

- **M.B. Azhari**, D. Han, S. Park, and D.H. Shim. "Robust Tightly-Coupled Filter-based Monocular Visual-Inertial State Estimation and Graph-based Evaluation for Autonomous Drone Racing," Under review at IEEE IROS 2026. [paper][video]
- **M.B. Azhari**, D. Han, Y. Je In, S. Park, and D.H. Shim. "Drift Corrected Monocular VIO and Perception-Aware Planning for Autonomous Drone Racing," Under review at Journal of Field Robotics 2026. [paper][video]
- **M.B. Azhari**, and D.H. Shim. "DINO-VO: A Feature-based Visual Odometry Leveraging a Visual Foundation Model," IEEE Robotics and Automation Letters, September 2025. (Presented at IEEE ICRA 2026) [paper]
- G. Kang, O. Guenes, S. Lee, **M.B. Azhari**, D.H. Shim. "SPIBOT: A Drone-Tethered Mobile Gripper for Robust Aerial Object Retrieval in Dynamic Environments," IEEE ICRA 2025. [paper] [video]
- B. Kim, **M.B. Azhari**, J. Park, and D.H. Shim. "An autonomous UAV system based on adaptive LiDAR Inertial Odometry for practical exploration in complex environments," Journal of Field Robotics, 2024. [paper] [video1] [video2]

EXPERIENCE

Graduate Student Researcher

Aug. 2021 – Present

Unmanned System Research Group, KAIST (Advisor: Prof. David Hyunchul Shim)

Daejeon, Korea

- Designed and deployed perception and state estimation systems for real-world autonomous UAVs across diverse missions including high-speed racing, maritime security, indoor reconnaissance, decontamination, and anti-drone UAVs.
- Developed tightly-coupled monocular VINS for autonomous drone racing (ADR-VINS), achieving 15 cm translational RMSE at speeds up to 20.9 m/s — deployed in A2RL Season 2.
- Led the development of UAV software stacks (perception, state-estimation, mission planner, and control) for a collaborative UAV-USV system enabling autonomous maritime intervention in GNSS-denied environments — achieved 2nd place among 30+ teams globally at MBZIRC Maritime Challenge.
- Developed full-stack autonomous indoor exploration UAV with adaptive LiDAR-inertial odometry, 3D path planning, exploration, and onboard object detection and 3D localization — won 1st Place at Army DroneBot Challenge and Top 2 at ADD Autonomous Drone Competition (2022).

Backend/Machine Learning Engineer

Aug. 2020 – Jul. 2021

XETIA.IO

Depok, Indonesia

- Developed microservices for user management, real-time chat, notifications, and AI model serving using Django, FastAPI, Redis, Celery, TensorFlow-serving, and Docker.

Research Assistant

Aug. 2020 – Jul. 2021

Computational Intelligence and Intelligent System Lab, University of Indonesia

Depok, Indonesia

- Conducted research in control system algorithm and computer vision; assisted undergraduate students' researches.

Technology Research and Development Intern

Jul. 2019 – Aug. 2019

Industri Kereta Api (State-own Rolling-stock Manufacturer)

Madiun, Indonesia

- Designed and simulated an Engine-Generator Control Unit and its electrical loads for an Electric-Diesel Train system.

HONORS & AWARDS

- 2026 – KAIST Challenge Award (Received for winning 3rd Place in A2RL Season 1)
- 2026 – 4th Place, A2RL Drone Championship Season 2 (Prize: USD 45,000.00)
- 2025 – 3rd Place, A2RL Autonomous Drone Racing Season 1 (Prize: USD 115,000)
- 2025 – KAIST Challenge Award (Received for winning 2nd Place in MBZIRC)
- 2024 – 2nd Place, Mohammed Bin Zayed International Robotics Challenge (MBZIRC)
- 2022 – Top 2, ADD Autonomous Drone Competition for Battelfield Environments
- 2022 – 1st Place, 5th Army Tiger DroneBot Challenge: Indoor Reconnaissance Drone

TECHNICAL SKILLS

Programming: C/C++, Python, C#, MATLAB

Frameworks & Tools: ROS/ROS2, PyTorch, OpenCV, Git, Docker

Robotics & Localization: Visual(-Inertial) Odometry/SLAM, LIDAR Odometry/SLAM, Sensor Fusion, Object Detection, Learning-based Methods, Autonomous Drones, and Other Robotic Stacks (Planning, Control, Exploration, etc).